Partial Differential Equations

An equation which involves more than one independent variable and one or more partial derivatives of the dependent variable w.r.t to them is called a partial differential equation.

The order of a partial differential equation is the order of the highest partial derivative appearing in the equation.

The degree of a partial differential equation is the highest power of higher order partial derivative occurring in the equation.

Examples:

1. $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = z \implies$ order = 1, degree = 1

2. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0 \implies$ order = 2, degree = 1

3. $\frac{\partial^2 z}{\partial x \partial y} = \left(\frac{\partial z}{\partial y}\right)^3 \implies$ order = 2, degree = 1

4. $\left(\frac{\partial^2 z}{\partial y^2}\right)^3 - \left(\frac{\partial z}{\partial y}\right)^3 \implies$ order = 2, degree = 1

5. $\left(\frac{\partial^2 z}{\partial x^2}\right)^3 - 3\left(\frac{\partial z}{\partial x}\right)^4 + \frac{\partial^2 z}{\partial x \partial y} = 5z \implies$ order = 2, degree = 1

Notations: If z is a function of two independent variables x and y, then

$$p = \frac{\partial z}{\partial x},\ q = \frac{\partial z}{\partial y},\ r = \frac{\partial^2 z}{\partial x^2},\ s = \frac{\partial^2 z}{\partial y^2},\ t = \frac{\partial^2 z}{\partial x \partial y}$$

Note:1 The number of variables in a PDE is the number of independent variables in that equation.

Note:2 If $u = f(x, y)$ where $x = \phi(t)$ and $y = \psi(t)$, then we can express u as a function of t alone by substituting the values of x and y in $f(x, y)$. Thus we can find the ordinary derivative $\frac{du}{dt}$ which is called the total derivative of u as

$$\frac{du}{dt} = \frac{\partial u}{\partial x}.\frac{dx}{dt} + \frac{\partial u}{\partial y}.\frac{dy}{dt} \quad \text{(chain rule)}$$

Note:3 If $u = f(x, y)$ where $x = \phi(s, t)$ and $y = \psi(s, t)$, then

$$\begin{aligned} \frac{\partial u}{\partial s} &= \frac{\partial u}{\partial x}.\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}.\frac{\partial y}{\partial s} \\ \frac{\partial u}{\partial t} &= \frac{\partial u}{\partial x}.\frac{\partial x}{\partial t} + \frac{\partial u}{\partial y}.\frac{\partial y}{\partial t} \end{aligned}$$

1.1 Formation of Partial Differential Equations

The partial differential equation can be formed either by the elimination of arbitrary constants or by the elimination of arbitrary functions.

If the number of arbitrary constants to be eliminated is equal to the number of independent variables, the partial differential equations that arise are of first order.

If the number of arbitrary constants to be eliminated is more than the number of independent variables, the partial differential equation obtained is of second or higher order.

If the partial differential equation is obtained by the elimination of arbitrary functions, then the order of the partial differential equation is equal to the number of arbitrary functions eliminated.

Question 1.1.1. *Form the partial differential equation by eliminating the arbitrary constants: (Obtain expression for the arbitrary constants in terms of partial derivatives* $(p,\ q,\ r,\ s,\ t$ *etc) and variables and substitute in the given equation.)*

(1) $z = (x-a)^2 + (y-b)^2$

(2) $z = ax + by + a^2 + b^2$

(3) $z = (x^2 + a)(y^2 + b)$

(4) $z = (x+a)^2 + (y+b)^2 + c^2$

(5) $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

(6) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

(7) $z = a \log\left[\frac{b(y-1)}{1-x}\right]$

(8) $(x-a)^2 + (y-b)^2 = z^2 \cot^2 \alpha$

(9) $z = a(x+y) + b(x-y) + abt + c$

Question 1.1.2. *Find the differential equation of all spheres of fixed radius having their centres in xy-plane.*

Question 1.1.3. *Find the differential equation of all planes which are at a constant distance* $'a'$ *from the origin.*

Question 1.1.4. *Form the partial differential equation by eliminating the arbitrary* *functions:*

(1) $z = f(\frac{y}{x})$

(2) $z = yf(\frac{y}{x})$

(3) $z = xg(y) + yf(x)$

(4) $x^2 + y^2 + z^2 = f(xy)$

(5) $z = f(x^2 - y^2)$

(6) $z = f(x + at) + g(x - at)$

(7) $z = f(\frac{xy}{z})$

(8) $xyz = \phi(x + y + z)$

(9) $z = f_1(x)f_2(y)$

(10) $z = y^2 + 2f(\frac{1}{x} + \log y)$

(11) $z = f_1(y + 2x) + f_2(y - 3x)$

(12) $z = x^2 f(y) + y^2 g(x)$

(13) $lx + my + nz = \phi(x^2 + y^2 + z^2)$

(14) $z = (x + y)f(x^2 - y^2)$

Formation of PDE by eliminating the arbitrary function F from $F(u, v) = 0$ where z is dependent on the independent variables x and y and u and v are functions of x, y and z.

1. Find $u_x,\ u_y, v_x,\ v_y$.
2. Solving $\begin{vmatrix} u_x & v_x \\ u_y & v_y \end{vmatrix} = 0$, we get the required PDE.

Question 1.1.5. *Form the PDE for* $f(x^2 + y^2, z - xy) = 0$.

Solution: *Consider* $f(x^2+y^2, z-xy)=0$. *Let* $u=x^2+y^2$, $v=z-xy$. *Then the given equation becomes* $f(u,v)=0$.*Now,*

$$u_x=2x,\ u_y=2y,\ v_x=p-y,\ v_y=q-x.$$

Consider $\begin{vmatrix} u_x & v_x \\ u_y & v_y \end{vmatrix}=0$

$\implies \begin{vmatrix} 2x & p-y \\ 2y & q-x \end{vmatrix}=0 \implies qx-py=x^2-y^2$ *is the required PDE.*

Question 1.1.6. *Form the PDE for* $f(xy+z^2, x+y+z)=0$.

Solutions of Partial Differential Equations

1.1.1 Equations solvable by direct integration

The problems of this type are done in two techniques - that of direct integration or by using ordinary differential equations. The point to be noted is that here, instead of the usual constants of integration, we use arbitrary functions of the variable that is held fixed.

Question 1.1.7. *Solve:* $\frac{\partial^2 z}{\partial y \partial x}=xy$.

Solution: *Given* $\frac{\partial^2 z}{\partial y \partial x}=xy$.

$\implies \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=xy$.

Keeping x *fixed and integrating w.r.t* y,

$$\frac{\partial z}{\partial x}=\frac{xy^2}{2}+f(x)$$

Now keeping y *fixed and integrating w.r.t* x,

$$z=\frac{x^2y^2}{4}+\int f(x)dx+g(y).$$

1.1.2 Some special types of non-linear first order PDE

Form I: $f(p, q) = 0$ i.e., equation contains only p and q (or x, y, z are absent) Assume that $p = a$ then $f(a, q) = 0$ Solving $q = \phi(a)$ Consider $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy = p\,dx + q\,dy$ $dz = a\,dx + \phi(a)dy$ Integrating $z = ax + \phi(a)y + c$ where a and c are arbitrary constants. Thus the complete solution is $z = ax + by + c$ where a, b satisfy the equation $f(a, b) = 0$ i.e., $b = \phi(a)$	**Form II: $f(z, p, q) = 0$** i.e., equation does not involve the independent variables x and y. Assume $q = ap$. Substituting q in the given equation $f(z, p, ap) = 0$ and solving for p, we get $p = \phi(z)$. Now $dz = pdx + qdy = pdx + ap\,dy$ $dz = p(dx + ady) = \phi(z)(dx + ady)$ Integrating $x + ay = \int \frac{dz}{\phi(z)} + b$ where a and b are two arbitrary constants.
Form III: $f(x, p) = g(y, q)$ i.e., x, p and y, q are separable. Assume $f(x, p) = g(y, q) = a$ = constant. Solving each equation for p and q, we get $p = f_1(x, a)$ and $q = g_1(y, a)$ Now $dz = pdx + qdy = f_1(x, a)dx + g_1(y, a)dy$ Integrating $z = \int f_1(x, a)dx + \int g_1(y, a)dy + b$.	**Form IV: Clairaut Equation:** **$z = px + qy + f(p, q)$** The complete solution of this equation is $z = ax + by + f(a, b)$ which is obtained by replacing p by a and q by b in the given clairaut equation.

1.1.3 Lagrange's linear equation

A partial differential equation of the form $Pp + Qq = R$ where $P,\ Q,\ R$ are functions of $x,\ y,\ z$ is called a Lagrange's equation and the method to solve it is called a Lagrange's Method. To find the complete solution of Lagrange's equation, we can use the following steps:

1. Form the auxiliary (OR subsidiary) equations: $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$.

2. Find any two independent solutions of the auxiliary equation, (say) $u = a$ and

$v = b$. This is usually done by treating the subsidiary equations as fractions.

There are two methods:

The method of grouping: Here, we group two out of the three fractions such that one variable is absent, or can be cancelled from both sides. This can now be solved as an ordinary differential equation.

The method of multipliers: Using 'multipliers' wherever necessary (multipliers are entries with which both numerator and denominator of a fraction are multiplied, so that its value doesn't change) and the principle of componendo dividendo (if $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$, then $each = \frac{la+mc+ne}{lb+md+nf}$) such that denominator $= 0$, we arrive at the relations $u = a$ and $v = b$.

3. Write the complete solution as $f(u, v) = 0$ or $u = \phi(v)$ or $v = \psi(u)$.

Question 1.1.8. *Solve* $(\frac{b-c}{a})yzp + (\frac{c-a}{b})zxq = (\frac{a-b}{c})xy.$

Solution: *This is a Lagrange's Equation of the form* $Pp + Qq = R$. *Here*

$$P = (\frac{b-c}{a})yz,\ Q = (\frac{c-a}{b})zx,\ R = (\frac{a-b}{c})xy.$$

The subsidiary equations are

$$\frac{dx}{(\frac{b-c}{a})yz} = \frac{dy}{(\frac{c-a}{b})zx} = \frac{dz}{(\frac{a-b}{c})xy}$$

We have

$$\frac{adx}{(b-c)yz} = \frac{bdy}{(c-a)zx} = \frac{cdz}{(a-b)xy}$$

By the method of grouping, $\frac{adx}{(b-c)yz} = \frac{bdy}{(c-a)zx}$

$\implies \frac{a}{(b-c)}xdx = \frac{b}{(c-a)}ydy$

Integrating, $\frac{a}{(b-c)}\frac{x^2}{2} = \frac{b}{(c-a)}\frac{y^2}{2} + \frac{k_1}{2}$

$$\implies (\frac{a}{b-c})x^2 - (\frac{b}{c-a})y^2 = k_1$$

Again, by the method of grouping, $\frac{bdy}{(c-a)zx} = \frac{cdz}{(a-b)xy}$

$\implies \frac{b}{(c-a)}ydy = \frac{c}{(a-b)}zdz$

Integrating, $\frac{b}{(c-a)}\frac{y^2}{2} - \frac{c}{(a-b)}\frac{z^2}{2} = \frac{k_2}{2}$

$$\implies (\frac{b}{c-a})y^2 - (\frac{c}{a-b})z^2 = k_2$$

Therefore the required solution is

$$\phi\left((\frac{a}{b-c})x^2 - (\frac{b}{c-a})y^2, (\frac{b}{c-a})y^2 - (\frac{c}{a-b})z^2\right) = 0.$$

1.1.4 Linear homogeneous partial differential equation with constant coefficient

A linear partial differential equation is a partial differential equation which contains partial derivatives of any order but none of them appears in degree higher than one. Examples: $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ is an example of a First-Order Linear PDE. $(x^2+y^2)\frac{\partial z}{\partial x} + \frac{\partial^2 z}{\partial x \partial y} - 3z = 0$ is a Second-Order Linear PDE whereas $(x^2+y^2)\frac{\partial z}{\partial x} + (\frac{\partial^2 z}{\partial x \partial y})^4 - 3z = 0$ is a Second-Order Non-Linear PDE.

If all partial derivatives in a linear partial differential equation are of same order, then it is called a homogeneous linear partial differential equation.

Example: $\frac{\partial^2 z}{\partial x^2} - 3\frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2} = 0$

A partial differential equation of the form

$$a_0\frac{\partial^n z}{\partial x^n} + a_1\frac{\partial^n z}{\partial x^{n-1}\partial y} + +a_2\frac{\partial^n z}{\partial x^{n-2}\partial y^2} + \ldots + a_n\frac{\partial^n z}{\partial y^n} = f(x,y) \tag{1.1}$$

where $a_0, a_1, \ldots, a_n$ are constants is called a homogeneous linear partial differential equation with constant coefficients.

We use the following notations: $\frac{\partial}{\partial x} = D$ and $\frac{\partial}{\partial y} = D'$. Then equation 1.1 can be written as

$$\begin{aligned}
a_0 D^n z + a_1 D^{n-1} D' z + +a_2 D^{n-2} D'^2 z + \ldots + a_n D'^n z &= f(x, y) \\
\text{or} \\
(a_0 D^n + a_1 D^{n-1} D' + +a_2 D^{n-2} D'^2 + \ldots + a_n D'^n) z &= f(x, y) \\
\text{or} \\
F(D, D') z &= f(x, y)
\end{aligned}$$

where $F(D, D') = (a_0 D^n + a_1 D^{n-1} D' + +a_2 D^{n-2} D'^2 + \ldots + a_n D'^n)$.

Working rule to find the Complimentary function:

1. Put the given equation in the standard form:

$$F(D, D') z = f(x, y) \tag{1.2}$$

   where $F(D, D') = (a_0 D^n + a_1 D^{n-1} D' + +a_2 D^{n-2} D'^2 + \ldots + a_n D'^n)$.

2. Replacing D by m and D' by 1 in $F(D, D')$, we obtain auxiliary equation (A.E.) as

$$a_0 m^n + a_1 m^{n-1} + +a_2 m^{n-2} + \ldots + a_n = 0. \tag{1.3}$$

3. Solve equation 1.3 for m.

4. If $m = m_1,\ m_2,\ \ldots,\ m_n$ are different roots,

$$z = f_1(y + m_1x) + f_2(y + m_2x) + \ldots + f_n(y + m_nx)$$

where $f_1,\ f_2,\ \ldots,\ f_n$ are arbitrary functions.

5. If a root m is repeated r times, *i.e.* $m_1 = m_2 = \ldots = m_r = m$,

$$\begin{aligned} z &= f_1(y + mx) + xf_2(y + mx) + x^2f_3(y + mx) + \ldots + x^{r-1}f_r(y + mx) + \\ &\quad f_{r+1}(y + m_{r+1}x) + \ldots + f_n(y + m_nx) \end{aligned}$$

where $f_1,\ f_2,\ \ldots,\ f_n$ are arbitrary functions.

Working rule to find the Particular integral:

Consider the equation $F(D, D')z = f(x, y)$, the particular integral is given as

$$z = \frac{1}{F(D, D')} f(x, y).$$

We find particular integrals for some specific forms of $f(x, y)$.

Case I: When $f(x, y) = e^{ax+by}$

$$[F(D, D')]^{-1} e^{ax+by} = \frac{1}{F(a, b)} e^{ax+by},$$

provided $F(a, b) \neq 0$.

In case $F(a, b) = 0$, we may follow the general method to be discussed as **Case IV**.

Case II: When $f(x, y) = \sin(ax + by)$ **or** $\cos(ax + by)$

$$[F(D^2, DD', D'^2)]^{-1} \sin(ax + by) = \frac{1}{F(-a^2, -ab, -b^2)} \sin(ax + by),$$

and

$$[F(D^2, DD', D'^2)]^{-1} \cos(ax + by) = \frac{1}{F(-a^2, -ab, -b^2)} \cos(ax + by),$$

provided $F(-a^2, -ab, -b^2) \neq 0$.

In case $F(-a^2, -ab, -b^2) = 0$, we may follow the general method to be discussed as **Case IV**.

Case III: When $f(x, y) = x^m y^n$**, or a polynomial in** $x,\ y$.

The particular integral is

$$z = [F(D, D')]^{-1} x^m y^n.$$

To evaluate it we expand $[F(D, D')]^{-1}$ as an infinite series in ascending powers of D or D', depending upon $m < n$ or $m > n$, and then operate on $x^m y^n$ term by term.

Case IV: The general method

When $f(x, y)$ is any function of x and y, under the assumption that $F(D, D')$ can be factorized linearly, consider

$$(D - mD')z = f(x, y).$$

Then

$$\frac{1}{(D - mD')} f(x, y) = \int f(x, c - mx) dx,$$

where c is to be replaced by $y + mx$ after integration.

1.1.5 Method of Separation of Variables

Method of separation of variables is a powerful tool to solve initial boundary value problem (IBVP) when PDE is linear and boundary conditions are homogeneous.

For a PDE in the function u of two independent variables x and y, assume that the required solution is separable, *i.e.*,

$$u(x,y) = X(x).Y(y) \tag{1.4}$$

where $X(x)$ is a function of x alone and $Y(y)$ is a function of y alone.

Then substitution of u from 1.4 and its derivatives reduces the PDE to the form

$$f(X, X', X'', \ldots) = g(Y, Y', Y'', \ldots) \tag{1.5}$$

which is separable in X and Y.

Since the LHS of 1.5 is a function of x alone and RHS of 1.5 is a function of y alone, then 1.5 must be equal to a common constant say k. Thus 1.5 reduces to

$$f(X, X', X'', \ldots) = k \tag{1.6}$$

$$g(Y, Y', Y'', \ldots) = k \tag{1.7}$$

Thus the determination of solution to PDE reduces to the determination of solutions to two ODE (with specific conditions).

From the expressions of $X(x)$ and $Y(y)$ so obtained we find expressions for $u(x,y)$, the desired solution.

Question 1.1.9. *Using method of separation of variables, solve* $3u_x + 2u_y = 0$ *where* $u(x,0) = 4e^{-x}$.

Solution: *Assume that* $u(x,y) = X(x).Y(y)$ *is a solution of given PDE. Differentiating* u *w.r.t* x *and* y*, we get* $u_x = X'Y$ *and* $u_y = XY'$ *so the given PDE* *reduces to*

$$\begin{aligned} 3u_x + 2u_y = 0 &\implies 3X'Y + 2XY' = 0 \\ &\implies 3X'Y = -2XY' \\ &\implies 3\frac{X'}{X} = -2\frac{Y'}{Y} = k \text{ (say)} \end{aligned}$$

Then

$$\begin{aligned} 3\frac{X'}{X} = k &\implies \frac{X'}{X} = \frac{k}{3} \\ &\implies \log X = \frac{k}{3}x + \log c_1 \\ &\implies \log(\frac{X}{c_1}) = \frac{k}{3}x \\ &\implies X = c_1 e^{\frac{k}{3}x} \end{aligned}$$

$$\begin{aligned} -2\frac{Y'}{Y} = k &\implies \frac{Y'}{Y} = -\frac{k}{2} \\ &\implies \log Y = \frac{-k}{2}y + \log c_2 \\ &\implies \log(\frac{Y}{c_2}) = \frac{-k}{2}y \\ &\implies Y = c_2 e^{\frac{-k}{2}y} \end{aligned}$$

Thus, solution of given PDE is given by

$$u(x,y) = X(x).Y(y) = [c_1 e^{\frac{k}{3}x}][c_2 e^{\frac{-k}{2}y}] = ce^{k(\frac{x}{3}-\frac{y}{2})}$$

Given $u(x,0) = 4e^{-x} \implies ce^{k(\frac{x}{3}-\frac{0}{2})} = 4e^{-x} \implies c = 4$ *and* $k = -3$.

Hence the required solution is $u(x,y) = 4e^{-3(\frac{x}{3}-\frac{y}{2})}$.

Question 1.1.10. *Solve* $u_{xx} - u_y = 0$ *by separation of variables.*

Solution: *Assume that* $u(x,y) = X(x).Y(y)$ *is a solution of given PDE.*

Differentiating u w.r.t x and y, we get $u_{xx} = X''Y$ *and* $u_y = XY'$ *so the PDE reduces to*

$$X''Y - XY' = 0 \implies X''Y = XY' \implies \frac{X''}{X} = \frac{Y'}{Y} = k$$

Solving $X'' - kX = 0$, *we have A.E* $m^2 - k = 0$, *solution is* $X(x) = c_1e^{\sqrt{k}x} + c_2e^{-\sqrt{k}x}$.

Solving $Y' - Y = k$, *we have* $Y(y) = c_3e^{ky}$.

Hence the solution is

$$u(x,y) = X(x).Y(y) = \left(c_1e^{\sqrt{k}x} + c_2e^{-\sqrt{k}x}\right)\left(c_3e^{ky}\right) = \left(Ae^{\sqrt{k}x} + Be^{-\sqrt{k}x}\right)\left(e^{ky}\right).$$

Note: To solve a second order ODE with constant coefficients: $y'' + ay' + by = 0$. Write the auxiliary equation A.E. by replacing y by 1, y' by m and y'' by m_2.

The general solution (G.S.) of the homogeneous D.E. is obtained depending on the nature of the two roots m_1 and m_2 of the auxiliary equation: $m^2 + am + b = 0$.

1. If $m_1$and m_2 are distinct and real, $y = c_1e^{m_1x} + c_2e^{m_2x}$.

2. If $m_1$and m_2 are real and equal $(m_1 = m_2 = m)$, $y = (c_1 + c_2x)e^{mx}$.

3. If $m_1 = a + ib, m_2 = a - ib$ (roots are complex conjugates), $y = e^{ax}[c_1 \cos bx + c_2 \cos bx]$.